

CSS Flexbox & Card Components

A Fun Coding Storybook & Practice Journey

What is CSS?

CSS stands for **Cascading Style Sheets**. It is the special language web developers use to add vibrant colours, fun fonts, custom sizes, breathing room spacing, and beautiful layouts to basic HTML pages.

 **The Clothes Analogy:** Think of HTML as the bare bones skeleton or plain white structural sketch of a webpage, and CSS as the amazing decorations! Just like a plain white outfit transforms into a spectacular festival outfit with colourful embroidery, bright buttons, and patterns, CSS transforms plain text HTML into something beautiful that you love to look at!

A Sneak Peek at a CSS Rule

```
.card {  
  background-color: white;  
  border-radius: 12px;  
  padding: 16px;  
}
```

What is Flexbox?

Flexbox (which is short for **Flexible Box Layout**) is a magic CSS tool that makes it super easy to arrange elements neatly in rows or columns. It distributes space between items and aligns them perfectly across your screen completely automatically!

 **The Tiffin Box Analogy:** The whole tiffin box (dabba) is the **Flex Container** — it holds everything together safely. Each separate section inside (where you put your rice, dal, and vegetables) is a **Flex Item** — neatly organised and perfectly fitted inside!

Element Role	Real-Life Analogy
Flex Container (The Parent)	A teacher in the school assembly providing guidelines on how to line up.
Flex Item (The Child)	The students standing beautifully and neatly following those instructions.

To wake up this layout superpower and turn on Flexbox, we simply apply the magic command `display: flex;` to the container element.

Key Flexbox Properties

The Direction Arrow (flex-direction)

This controls whether your items choose to line up side-by-side or stack up on top of each other like a tower.

- `row`: Items travel from LEFT to RIGHT. This is the default setting, just like seats in a school bus!
- `column`: Items travel from TOP to BOTTOM. This stacks items neatly, just like the floors in a tall building!

Left-to-Right Spacing (justify-content)

This property controls how your items spread out or share horizontal space inside their parent container.

- `center`: Everything packs tightly together right in the middle of the screen.
- `space-between`: Items push all the way out to the edges, leaving perfectly equal pockets of space directly between them.

Up-to-Down Alignment (align-items)

This controls how your items match up vertically with each other.

- `center`: Everything balances perfectly in the middle vertically.
- `stretch`: Pulls and stretches items all the way from the absolute top to the very bottom to fill up the space.

| What is a Card Component?

A **card component** is a rectangular digital box that displays a unique piece of information beautifully. It usually contains an eye-catching image at the top, a bold title, a short descriptive text paragraph, and an action button at the bottom. You interact with them every single day on your favourite apps, such as food cards on Zomato, video thumbnails on YouTube, and toy product listings on Flipkart!



Practice Worksheet

Show off your coding superpowers! Answer the practice questions below.

Question 1: Fill in the Blank

CSS stands for Cascading Sheets.

Question 2: Multiple Choice

If HTML is like a plain skeleton, what is CSS like?

- A) The engine of a car
- B) Beautiful decorations, colors, and festive outfits
- C) The bones inside an animal
- D) A computer mouse

Question 3: True or False

True / False: To activate Flexbox layout, you must add the rule `display: flex;` to the Flex Container (the parent).

Question 4: Match the Analogy

In our Tiffin Box analogy, if the whole tiffin box is the **Flex Container**, what are the delicious food compartments inside?

Answer:

Question 5: Multiple Choice

Which property changes whether items are arranged in a horizontal row or a vertical column?

- A) `background-color`
- B) `border-radius`
- C) `flex-direction`
- D) `gap`

Question 6: Drawing & Code Connection

You want your custom cards to look like floors in a building, stacked from top to bottom. Which code value should you use?

`flex-direction: .....`;

Question 7: Multiple Choice

Which value of `justify-content` spreads out items evenly so that the first item is at the start and the last item is at the absolute end?

- A) `flex-start`
- B) `center`
- C) `space-between`
- D) `column`

Question 8: Fill in the Blank

To make a card's corners nice, round, and friendly instead of sharp, we use the CSS property called `border-.....`.

Question 9: Identify the Element

Name three things that you usually find inside a standard web **card component** (like a product card on Flipkart or a restaurant card on Zomato):

1. 2. 3.
.....

Question 10: Multiple Choice

If you want to add 16px of breathing room or space directly *between* two cards, which property is the easiest to use?

- A) `gap: 16px;`
- B) `display: block;`
- C) `color: white;`
- D) `font-size: 16px;`

Question 11: Application Question

A teacher wants her students' photo cards to stay in the exact center from left to right on the website. Which horizontal alignment property should she apply to the container?

`justify-content: .....`;

Question 12: Creative Challenge

When you hover your mouse pointer over a card, you want it to lift up smoothly. What special CSS state word do we use after a colon (e.g., `.card:...`)?

- A) `click`
- B) `hover`
- C) `lift`
- D) `flex`